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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 13

Fungus Resistance

Important Notice

Information contained in these sections is pertinent to all test procedures described in the other sections of this document. Further, Appendix A is applicable for identifying environmental tests performed.

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13.0 Fungus Resistance

13.1 Purpose of the Test

These tests determine whether equipment material is adversely affected by fungi under conditions favorable for their development, namely, high humidity, warm atmosphere and presence of inorganic salts.

Notes:

A. Fungi proximity to other materials, exposure to daily susceptible contaminants such as fluids during routine operation and maintenance, or equipment exposure to solar actinic effects - may break molecular bonds and reduce the item to sub-compositions which may be fungus nutrients.

B. This test shall not be conducted after Salt Spray or Sand and Dust. A heavy concentration of salt may effect the fungal growth, and sand and dust can provide nutrients, which could compromise the validity of this test (see Subsection 3.2, "Order of Tests").

13.2 General Effects

Typical problems caused by fungi growing on equipment are:

a. Microorganisms digest organic materials as a normal metabolic process, thus degrading the substrate, reducing the surface tension and increasing moisture penetration.

b. Enzymes and organic acids, produced during metabolism, diffuse out of the cells and onto the substrate and cause metal corrosion, glass etching, hardening of grease and other physical and chemical changes to the substrates.

c. The physical presence of microorganisms produces living bridges across components that may result in electrical failures.

d. The physical presence of fungi can also cause health problems and produce aesthetically unpleasant situations in which users will reject using the equipment.

The detrimental effects of fungal growth are summarized as follows:

a. Direct attack on materials. Nonresistant materials are susceptible to direct attack as fungus breaks these materials down and uses them as nutrients. This results in deterioration affecting the physical properties of the material. Examples of nonresistant materials are:

(1) Natural material. Products of natural origin (carbon based) are most susceptible to this attack.

(a) Cellulose materials (e.g., wood, paper, natural fiber textiles, and cordage).

(b) Animal- and vegetable-based adhesives.

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(c) Grease, oils, and many hydrocarbons.
(d) Leather.
(2) Synthetic materials.
(a) PVC formulations (e.g., those plasticized with fatty acid esters).
(b) Certain polyurethanes (e.g., polyesters and some polyether).
(c) Plastics that contain organic fillers of laminating materials.
(d) Paints and varnishes that contain susceptible constituents.

b. Indirect attack on materials. Damage to fungus-resistant materials results from indirect attack when:

(1) Fungal growth on surface deposits of dust, grease, perspiration, and other contaminants (that find their way onto materiel during manufacture or accumulate during service) causes damage to the underlying material, even though that material may be resistant to direct attack.
(2) Metabolic waste products (i.e., organic acids) excreted by fungus cause corrosion of metals, etching of glass, or staining or degrading of plastics and other materials.
(3) The acidic waste products of fungus on adjacent materials that are susceptible to direct attack come in contact with the resistant materials.

13.3 Categories of Equipment

Category F

Equipment that is installed in an environment where it will be exposed to severe fungus contamination is identified as Category F and shall be subjected to the fungus resistance test. If all materials used in the construction of the equipment can be shown to be non-nutrients for the growth of fungi, either through their composition or through previous testing, this test is not required. If non-nutrient material certification is u tilized for this verification, this fact shall be declared on the Environmental Qualification Form (see Appendix A).

13.4 Apparatus

The apparatus required to conduct this test consists of chambers or cabinets together with auxiliary instrumentation capable of maintaining the specified condition of temperature and humidity. Provisions shall be made to prevent condensation from dripping on the test item. There shall be free circulation of air around the test item and the contact area of fixtures supporting the test item shall be kept to a minimum. When forced air is employed, the flow should not exceed one meter per second over the surface of the test specimen.

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13.5 Test Procedures

13.5.1 Preparation of Mineral-Salts Solution

The solution shall contain the following:

Potassium dihydrogen orthophosphate	0.7 g
Potassium monohydrogen orthophosphate	0.7 g
Magnesium sulfate heptahydrate	0.7 g
Ammonium nitrate	1.0 g
Sodium chloride	0.005 g
Ferrous sulfate heptahydrate	0.002 g
Zinc sulfate heptahydrate	0.002 g
Manganous sulfate monohydrate	0.001 g
Distilled Water	1000 ml

Sterilize the mineral salts solution by autoclaving at 121 degrees C for 20 minutes. Adjust the pH of the solution by the addition of 0.01 normal solution of sodium hydroxide so that after sterilization the pH level is between 6.0 and 6.5. Prepare sufficient salt solutions for the required tests.

13.5.1.1 Purity of Reagents

Reagent grade chemicals shall be used in all tests. Unless otherwise specified, it is intended that all reagents shall conform to the specification of the Committee on Analytical Reagents of the American Chemical Society, where such specifications are available.

13.5.1.2 Purity of Water

Unless otherwise specified, references to water shall be understood to mean distilled water or water of equal purity.

13.5.2 Preparation of Mixed Spore Suspension

The following test fungi shall be used:

Fungi	ATCC[1]
Aspergillus niger	9642
Aspergillus flavus	9643
Aspergillus versicolor	11730
Penicillium funiculosum	11797
Chaetomium globosum	6205

[1] American Type Culture Collection, 12301 Parklawn Drive, Rockville, Maryland, 20852.

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Maintain cultures of these fungi separately on an appropriate medium such as potato dextrose agar. However, the culture of chaetomium globosum shall be cultured on strips of filter paper on the surface of mineral salts agar. (Mineral salts agar is identical to the mineral salts solution described in paragraph 13.5.1, but contains in addition 15.0 g of agar/liter.) The stock cultures may be kept for not more than four months at 6 ±4 degrees C, at which time subcultures shall be made and new stocks shall be selected from the subcultures. If genetic or physiological changes occur, obtain new cultures as specified above. Subcultures used for preparing new stock cultures or the spore suspension shall be incubated at 30 degrees C for seven to ten days. Prepare a spore suspension of each of the five fungi by pouring into one subculture of each fungus a 10 ml portion of a sterile solution containing 0.05 g/liter of a non-toxic wetting agent such as sodium dioctyl sulfosuccinate or sodium lauryl sulfate. Use a sterile platinum or nichrome inoculating wire to scrape gently the surface growth from the culture of the test organism. Pour the spore charge into a sterile 125 ml glass-stoppered Erlenmeyer flask containing 45 ml of sterile water and 10 to 15 solid glass beads, five millimeters in diameter. Shake the flask vigorously to liberate the spores from the fruiting bodies and to break the spore clumps. Filter the dispersed fungal spore suspension through a six millimeter layer of glass wool, contained in a glass funnel, into a sterile flask. This process should remove large mycelial fragments and clumps of agar that could interfere with the spraying process. Centrifuge the filtered spore suspension aseptically and discard the supernatant. Resuspend the residue in 50 ml of sterile water and centrifuge. Wash the spores obtained from each of the fungi in this manner three times. Dilute the final washed residue with sterile mineral-salts solution in such a manner that the resultant spore suspension shall contain 1,000,000 ±200,000 spores/milliliter as determined with a counting chamber. Repeat this operation for each organism used in the test and blend equal volumes of the resultant spore suspension. The spore suspension may be prepared fresh each day and maintained at 6 ±4 degrees C not more than fourteen days.

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13.5.3 Viability of Inoculum Control

With each daily group of tests, place each of three pieces of sterilized filter paper, 2.54 cm square on hardened mineral-salts agar in separate Petri dishes. Inoculate these with the spore suspension by spraying the suspension from a sterilized atomizer[1] until initiation of droplet coalescence. Incubate these at 30 degrees C at a relative humidity not less than 85 percent, and examine them after seven days of incubation. There shall be copious growth on all three of the filter paper control specimens. Absence of such growth requires repetition of the test.

13.5.4 Control Items

In addition to the viability of inoculum control, known susceptible substrates shall be inoculated along with the test item to insure that proper conditions are present in the incubation chamber to promote fungus growth. The control items shall consist of (unbleached) cotton duck 234 g strips that are 3.2 cm wide, that have been dipped into a solution containing 10% glycerol, 0.1% potassium dihydrogen orthophosphate, 0.1% ammonium nitrate, 0.025% magnesium sulfate, and 0.05% yeast extract (pH level 5.3), and from which the excess liquid has been removed. The strips should be hung to air dry before being inoculated and placed into the chamber.

13.5.5 Inoculation of Test Control

a. Mount the test and control items on suitable fixtures or suspend from hangers.

b. Precondition the chamber and its contents at 30 degrees C and 97 ±2% relative humidity for at least four hours.

c. Inoculate the test and control items with the mixed fungus spore suspension (paragraph 13.5.2) by spraying it on the test and control items in the form of a fine mist from a previously sterilized atomizer or nebulizer. In spraying the test and control items, care should be taken to cover all surfaces. If the surfaces are non-wetting, spray until initiation of droplet coalescence. Incubation is to be started immediately following the inoculation.

13.5.6 Incubation

a. Maintain the test chamber at 30 degrees C and 97 ±2% relative humidity (minimum) during the life of the test. Keep the test chamber closed during the incubation period, except during inspection or for addition of other test items.

b. After seven days, inspect the growth on the control items to be assured that the environmental conditions are suitable for growth. If inspection reveals that the environmental conditions are unsuitable for growth, the entire test shall be repeated.

c. If the control items show satisfactory fungus growth, continue the test for a period of 28 days from the time of inoculation or as specified in the equipment specification.

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[1] An atomizer capable of providing 15,000 ±3,000 spores/cm^2.

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13.5.7 Inspection

At the end of the incubation period, inspect the test item immediately. If possible, inspect the item within the chamber. If the inspection is not completed in eight hours, return the test item to the humid environment for a minimum of twelve hours. Except for hermetically sealed equipment, open the equipment enclosure and examine both the interior and exterior for evidence of deterioration. The equipment shall then be tested to DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

13.5.7.1 Analysis of Results.

The following information is provided to assist in the evaluation of the test results.

a. Any fungal growth on the test item must be analyzed to determine the species, and if the growth is on the test item material(s) or on contaminants.

b. Any fungal growth on the test item material(s), whether from the inoculums or other sources, must be evaluated by qualified personnel for:

(1) The extent of growth on susceptible components or materials. Any growth must be completely described.

(2) The immediate effect that the growth has on the physical characteristics of the materiel.

(3) The long-range effect that the growth could have on the materiel.

(4) The specific material (nutrient(s)) supporting the growth.

c. Evaluate human factors effects (including health risks).

13.5.8 Precautions

The fungi specified for this test are not normally considered a serious hazard for human handling. It is possible for an individual to be allergic to one of them, and for this reason it is wise to exercise care when performing the test. Surgical gloves may be worn to protect the hands, and care should be taken not to splash the suspension on other areas of the skin or on clothes. Additionally, an advisory prior to fungus inoculation should be issued instructing personnel possibly entering the vicinity where fungus spray inoculations or test preparations are taking place, to wear surgical masks and gloves as precautions to prevent adverse health reactions from fungus exposure.

It is also possible, during the incubation period in the test chamber, for a foreign spore, present as an unintentional intruder, to develop; some of these fungi thus present as native to some testing locations, may be injurious to the human system. For this reason there is a possibility that the specimen after exposure may be a hazard, and it should be handled with care.

The greatest danger, if some hazardous foreign spore is present on exposed specimens, is that small, dry, detached particles may become airborne and be carried into the lungs. This is only likely to happen after the specimen has dried out. If the specimen is carried quickly from the test chamber to a normal chemical fume hood before it has time to dry,

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the flow of air does not reach the operator and detached fragments cannot enter the nasal passages.

Detached portions of growth may be so small that no protection is offered by wearing a gauze mask and only a special respirator for sub-micron particles is effective. The use of a fume hood as suggested above, however, is considered an adequate precaution when performing this test.

Where the test location may contain such a harmful fungus, vestiges of it may remain in the test chamber and present a similar danger when it is being cleansed. High temperature steam, the preferred cleansing treatment, will render the chamber completely harmless. Where, however, fumigation with propylene oxide is adopted, it will be noted that fumigation prior to washing will ensure that all residues washed from the chamber are completely harmless.

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